Fiscal Year:	FY 2025	**Task Last Updated:**	FY 04/14/2025
PI Name:	Mehta, Satish Ph.D.		
Project Title:	Varicella Zoster Virus Shedding After Antiviral Drug (Valacyclovir) Treatment in Antarctic Expeditioners		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	HUMAN RESEARCH--Biomedical countermeasures		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HHC**:Human Health Countermeasures		
Human Research Program Risks:	(1) **Immune**:Risk of In Mission Impacts, Adverse Health Events or Long-Term Health Impacts due to Altered Immune Response		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Houston	**State:**	TX
Zip Code:	77058-2720	**Congressional District:**	36
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017 HERO 80JSC017N0001-Crew Health and Performance (FLAGSHIP1, OMNIBUS). Appendix A-Flagship1, Appendix B-Omnibus
Start Date:	07/10/2018	**End Date:**	09/30/2025
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Brocato, Becky	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/30/2025 to allow for publication of manuscripts, per T. Sirmons/HRP JSC (Ed., 4/11/25) NOTE: Final report date changed to 09/30/2024 per T. Sirmons/HRP JSC (Ed., 4/29/24) NOTE: End date changed to 10/1/2023 per T. Sirmons/HRP JSC (Ed., 2/23/23) NOTE: End date changed to 10/1/2022 per L. Singh/HRP JSC (Ed., 4/8/21)		
Key Personnel Changes/Previous PI:	April 14, 2025--One Ph.D. student completed her degree during this period. Additional Note: Duane Pierson is no longer at NASA, and has been removed from the list of Co-Investigators (Ed., 4/15/25).		
COI Name (Institution):	Crucian, Brian Ph.D. (NASA Johnson Space Center) Locke, James M.D. (NASA Johnson Space Center)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			

Performance Goal Text:	
Task Description:	Previous spaceflight studies indicate that reactivation of varicella zoster virus (VZV), particularly during longer duration spaceflights, can potentially lead to clinical disease including zoster, chronic neuropathic pain, vision loss, stroke, and cognitive impairment. Furthermore, continued viral shedding after spaceflight may cause clinical disease in crew contacts including uninfected or immunocompromised individuals, as well as newborn infants. Thus, it is essential to develop spaceflight countermeasures to prevent VZV reactivation and ensure the health of the crew, as well as the health of their contacts upon return. One such countermeasure is prophylactic administration of an antiviral drug (valacyclovir) against VZV. In order to determine the effectiveness of this countermeasure using a large population, we propose to study VZV shedding in Antarctic expeditioners who have similar patterns of VZV DNA shedding in saliva as astronauts. Countermeasure efficacy of the antiviral drug will be determined by measuring VZV reactivation and shedding in saliva as well as measuring the physiological stress biomarkers (cortisol, DHEA, and salivary amylase) and immune markers (inflammatory cytokines) before, during, and after the winter-over period. The proposed research team has extensive experience in ground-based studies including studies conducted in Antarctica, Aquarius undersea habitat, and artificial gravity, as well as the coordination and conduct of complex multi-laboratory studies. In addition, the research team has proven expertise and experience in immunology, virology, and medical expertise working with infectious diseases and spaceflight subjects. This proposal addresses the need for developing and validating countermeasures as identified in the new NASA Research Announcement (NRA) 80JSC017N0001-OMNIBUS NASA HERO Omnibus Opportunity.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Our studies demonstrated that reactivation of VZV, particularly during longer duration spaceflight, can potentially lead to clinical disease including zoster, chronic neuropathic pain, vision loss and cognitive impairment. Furthermore, continued viral shedding post-spaceflight may cause clinical disease in crew contacts including uninfected or immunocompromised individuals, as well as newborn infants. Thus, it is essential to develop spaceflight countermeasures to prevent VZV reactivation and ensure the health of the crew, as well as the health of their contacts upon return. One such countermeasure is prophylactic administration of an antiviral drug (valacyclovir) against VZV. In order to determine the effectiveness of this countermeasure with a relatively large population, we proposed to study VZV shedding in Antarctic expeditioners who have had similar patterns of VZV DNA shedding in saliva as astronauts. The results were very encouraging. We we found a significant decrease in the viral shedding (both EBV and HSV1) in subjects after prophylactic administration of Valtrex as compared to placebo during the winterover at the South Pole station as well as MuMurdo station. A 21-fold decrease in average EBV viral load (copies/ml saliva) was observed in the treatment group (antiviral acyclovir administration) than the placebo group at both stations. South Pole Station had a 280-fold decrease, while McMurdo Station had a 4-fold decrease in viral load. For HSV-1, the reduction was 5-fold at both stations. When parsing out each location, South Pole had a 3-fold increase in viral load and McMurdo had an 8.5-fold decrease, respectively. Interestingly, there was no VZV shedding in the saliva of expeditioners from both groups, so the effect of Valacyclovir on VZV reactivation and shedding cannot be inferred in this case. These. As viral reactivation is related with robust immune system of the host, it would be very critical to identify the immune status of these individual with and without antiviral intervention. Unfortunately, we could not study any of the immune markers in this study except for a limited panel of saliva cytokines (specifically 13) due to sampling constraints for this specific study, as well as budgetary and time concerns [Omnibus is one year proposal]. Our laboratory has previously determined both plasma and saliva cytokine alterations in ISS astronauts.
Task Progress:	VZV ANTIVIRALS AUGMENT Data Synopsis: A total of 1046 saliva samples were analyzed from 32 subjects enrolled in the Varicella Zoster Virus (VZV) Antivirals in Antarctica study. Nominally, five (5) passive drool saliva samples were collected at the beginning of every month for 6 (McMurdo) or 8 (South Pole) months. All samples were stored frozen until batch processing. In the already published paper, these samples were then thawed, aliquoted, and processed for viral (EBV, HSV1, and VZV) DNA by real-time polymerase chain reaction (PCR), stress hormone (cortisol, DHEA, and alpha-amylase), and cytokine analysis. The extra supernatant from these samples were refrozen and saved for this augment. The remaining saliva supernatant was used to analyze salivary C-reactive protein (CRP) and IgA, IgG1, IgG2, IgG3, IgG4, and IgM. Summary of Results: Nearly every sample was analyzed, but some samples were not analyzed due to too little volume resulting from previous use of the supernatant. In total, 624/625 for placebo and 420/429 for valacyclovir were able to be analyzed for this augment. CRP (pg/mL) was significantly less in the valacyclovir group (660 ± 36) compared to placebo (1472 ± 39) throughout the mission ($p < 0.0001$). This significance was observed similarly at both the South Pole and McMurdo stations. Significantly less CRP was observed between groups at every monthly timepoint except October (South Pole only). When parsing out the stations, however, South Pole did not begin to have a significant separation between the groups until May, which carried over all the way to September. March and April were not significant, but the valacyclovir group still had less CRP than the placebo group in both months. CRP was significantly less for valacyclovir at every month for McMurdo. No significant difference was observed in IgA (ng/mL) between the groups (placebo 3666 ± 120 vs. valacyclovir 3689 ± 126; $p = 0.8948$). With combined data from both stations, IgA was significantly increased in the valacyclovir group as compared to placebo in March (placebo 2968 ± 217 vs. valacyclovir 4167 ± 337). Interestingly, IgA was significantly less at the South Pole station ($p < 0.05$) but significantly more at the McMurdo station ($p < 0.0001$). IgA was significantly less for valacyclovir in September at the South Pole station but was significantly increased at McMurdo in both March and July. No significant difference was observed in IgM (ng/mL) between the groups when combining both station data either overall or separated by month. South Pole station, however, showed a significant reduction in IgM overall (placebo 2019 ± 100 vs. valacyclovir 1603 ± 101; $p < 0.01$). When separating out by each month and looking at placebo longitudinally, there was a significant increase in IgM in September compared to both March and April. No significant differences were seen at the McMurdo station. IgG1 (ng/mL) was significantly less in the valacyclovir group (6469 ± 410) compared to placebo (8780 ± 411)

	throughout the mission ($p < 0.0001$). This result was exclusively driven by the McMurdo station and not South Pole. No significant difference was observed in IgG2 (ng/mL) between the groups (placebo 26311 ± 1396 vs. valacyclovir 23348 ± 1186; $p = 0.1085$). Interestingly, IgG2 was significantly more at the South Pole station ($p < 0.05$), but significantly less at the McMurdo station ($p < 0.001$). No significant difference was observed at any given month throughout the campaign. IgG3 (ng/mL) was significantly less in the valacyclovir group (171.9 ± 6.417) compared to placebo (190.7 ± 5.825) throughout the mission ($p < 0.05$). No differences were seen at either station or at any given month. IgG4 (ng/mL) was significantly less in the valacyclovir group (380.2 ± 10.25) compared to placebo (446.4 ± 9.137) throughout the mission ($p < 0.0001$). This result was mostly driven by both McMurdo (difference between means -61.35 ± 12.41; $p < 0.0001$) and South Pole (difference between means -51.69 ± 20.36; $p < 0.05$). However, it must be noted that most of the data (65%) for the IgG4 analyte was outside the range of the detection limit of the assay (673 above, 2 below, 371 within). When removing these data points outside the range, there was basically no change between the groups.
Bibliography Type:	Description: (Last Updated: 04/15/2025)
Abstracts for Journals and Proceedings	Douglass DM, Mehta SK, Nelman-Gonzalez M, Gutierrez CL, Lopez SB, Crucian BE. "Prophylactic antiviral treatment dampens viral induced immune stress in Antarctic expeditioners during winterover." 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators' Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025